
Dollars, Debt, and Destiny: Marfrig's Dilemma in a Changing Currency World

As US trade policy disrupted the world monetary system in 2025, Marfrig Global Foods (Marfrig) CFO Tang David began to rethink the Brazilian company's financial strategy. Executives were finalizing a \$500 million bond issuance with a 30-year maturity and pricing a three-year international export contract, deals that the firm often completed in US dollars. Marfrig conducted business across many countries, but it regularly chose to make transactions in dollars because the US currency typically provided stability and security. The US dollar had served as the preeminent currency for global transactions for decades, but David wondered about the future of its long-held status. The value of the dollar had begun declining early that year, and the fall continued in April when the United States announced the imposition of extensive tariffs, creating uncertainty in the international sector. Aside from tariff policy, David pondered other factors that could affect the dollar's worldwide position, including the United States' growing government debt. Would the lack of clarity regarding US trade policy or the direction of US fiscal policy cause people to lose confidence in US institutions and therefore the US dollar? The weakening of the dollar might simply represent a temporary aberration, but it could also signal a more permanent shift in the global order. Was it still safe for Marfrig to anchor in US dollars?

International Monetary Systems and the Rise of the Dollar

Over time, various currencies had dominated international trade and finance. When the Republic of Florence (in what became Italy) developed into a commercial and banking hub in the 13th century, the Florentine florin (initially a gold coin and later a silver coin) became one of the first currencies to reign supreme, arising as the primary medium for large purchases across Europe and different parts of the Mediterranean region.¹ Other currencies eventually overtook the florin as the preferred means for cross-border transactions, with the British pound gaining wide acceptance by the 19th century, when Great Britain acted as a global financial leader.² The pound lost its worldwide standing after World War I (1914–18) and World War II (1939–45) ravaged Europe, while the relative strength of the US economy and US financial markets propelled the US dollar to dominant-currency status.³ (See **Exhibit 1** for a timeline of dominant currencies.)

¹ Thomas Hockenhull, "Florin a Coin with Roots in Florence, Global Legacy," *Coin World*, November 17, 2014, <https://www.coinworld.com/news/world-coins/gold-florin-coin-world-international-coins-florence-italy-numismatics-history-collecting-hobby-349230.html> (accessed Jan. 15, 2026).

² Kit Dawnay, "A History of Sterling," *The Telegraph*, October 8, 2001, <https://www.telegraph.co.uk/news/1399693/A-history-of-sterling.html> (accessed Jan. 15, 2026).

³ Some research suggested that the British pound and US dollar both served as dominant reserve currencies during the interwar period. See Barry Eichengreen and Marc Flandreau, "The Rise and Fall of the Dollar, or When Did the Dollar Replace Sterling as the Leading Reserve Currency?" (working paper 14154, National Bureau of Economic Research, Cambridge, MA, 2008), https://www.nber.org/system/files/working_papers/w14154/w14154.pdf (accessed Jan. 15, 2026).

This public-sourced case was prepared by Felipe Saffie, Assistant Professor of Business Administration, and Bryan Harrison, Senior Case Researcher. Tang David is a fictional character created for pedagogical purposes. It was written as a basis for class discussion rather than to illustrate effective or ineffective handling of an administrative situation. Copyright © 2026 by the University of Virginia Darden School Foundation, Charlottesville, VA. All rights reserved. To order copies, send an email to sales@ardenbusinesspublishing.com. No part of this publication may be reproduced, stored in a retrieval system, used in a spreadsheet, or transmitted in any form or by any means—electronic, mechanical, photocopying, recording, or otherwise—without the permission of the Darden School Foundation. This publication is protected by copyright and may not be uploaded in whole or part to any AI, large language model, or similar system, or to any related training database. Our goal is to publish materials of the highest quality, so please submit any errata to editorial@ardenbusinesspublishing.com.

As preferences for certain currencies shifted throughout the centuries, the monetary systems that governed international trade and finance evolved as well. While many early commercial transactions involved the use of both gold and silver, a global exchange system linked more directly to gold emerged in the late 19th century. Under the gold standard, a country defined the value of its currency in terms of gold to establish a fixed exchange rate; on demand, governments would swap their gold reserves for the domestic currency. This system served key trading countries until World War I, when some countries needed to print large sums of money to help finance the war, impairing their ability to convert domestic currency into gold and forcing them to abandon the traditional gold standard.⁴ Although major countries returned to a gold standard following the war, the system broke down during the Great Depression in the 1930s, as many countries sought to protect their industries with unilateral policies that curtailed trade, including competitive devaluations, higher tariffs, and restrictive trade blocs.⁵

Near the end of World War II, international leaders developed a revised monetary structure intended to offer more flexibility than prior systems and to inspire global cooperation for postwar economic rebuilding.⁶ With large parts of the globe devastated from war, the United States began to arise as the world's preeminent economic power, and as a result, the new international monetary system relied on the robustness of the US economy. At a conference in Bretton Woods, New Hampshire, in 1944, delegates from 44 countries agreed to establish the International Monetary Fund (IMF),⁷ which would administer a fixed-exchange-rate system that tied gold convertibility to the US dollar.⁸ Specifically, countries would peg their currencies to the dollar (within a 1% band), and the US government could convert dollars held by official foreign institutions into gold at the set price of \$35 per ounce.⁹ By the time the system became functional in the late 1950s, both international trade conditions and the economies of war-ravaged countries had generally improved, but the United States faced economic headwinds.¹⁰

While world leaders initially worried about a dollar shortage, the new system soon allowed some countries to accumulate large levels of dollar reserves. As dollars circulated, US gold reserves could no longer cover them, and the dollar became overvalued: Its exchange rate exceeded its purchasing power, making US goods more expensive than foreign goods.¹¹ (See **Appendix 1** for a model of the foreign exchange market for US dollars.) US officials feared that the strong dollar, along with speculation in currency markets, undermined US

⁴ "Money Matters: An IMF Exhibit – The Importance of Global Cooperation: Conflict and Cooperation (1871–1944), Part 1: The Golden Era," IMF, https://www.imf.org/external/np/exr/center/mm/eng/mm_cc_01.htm; "Money Matters: An IMF Exhibit – The Importance of Global Cooperation: Conflict and Cooperation (1871–1944), Part 2: Meltdown," IMF, https://www.imf.org/external/np/exr/center/mm/eng/mm_cc_02.htm; "Money Matters: An IMF Exhibit – The Importance of Global Cooperation: Conflict and Cooperation (1871–1944), Part 3: Cost of the World War," IMF, https://www.imf.org/external/np/exr/center/mm/eng/cc_sub_2.htm (both accessed Jan. 15, 2026).

⁵ "Bretton Woods-GATT, 1941–1947," in "Milestones in the History of U.S. Foreign Relations," US Office of the Historian, <https://history.state.gov/milestones/1937-1945/bretton-woods> (accessed Jan. 15, 2026).

⁶ Sandra Kollen Ghizoni, "Creation of the Bretton Woods System," Federal Reserve History, November 22, 2013, <https://www.federalreservehistory.org/essays/bretton-woods-created> (accessed Jan. 15, 2026).

⁷ For the formal agreement, see "Articles of Agreement: International Monetary Fund and International Bank for Reconstruction and Development," United Nations Monetary and Financial Conference, July 1–22, 1944, https://fraser.stlouisfed.org/files/docs/historical/martin/17_07_19440701.pdf (accessed Jan. 15, 2026).

⁸ The IMF also served to provide financial support to certain countries. In addition to the IMF, conference attendees approved the creation of the International Bank for Reconstruction and Development (IBRD) (World Bank), which would supply financial aid for war-torn countries and assistance for economic advancement in less-developed nations.

⁹ The price of gold in terms of dollars was set by executive proclamation in 1934. See "White House Statement on Proclamation 2072," White House press release, in the American Presidency Project, University of California, Santa Barbara, January 31, 1934, <https://www.presidency.ucsb.edu/documents/white-house-statement-proclamation-2072> (accessed Jan. 15, 2026).

The United States amassed a significant amount of gold during the World War II period and held 70% of the gold reserves in the world by 1947. See "Money Matters: An IMF Exhibit – The Importance of Global Cooperation: Destruction and Reconstruction (1945–1958), Part 1: The Post War World," IMF, https://www.imf.org/external/np/exr/center/mm/eng/mm_dr_01.htm (accessed Jan. 15, 2026).

¹⁰ "Money Matters: An IMF Exhibit – The Importance of Global Cooperation: Destruction and Reconstruction (1945–1958), Part 6: Economic Miracles in the 1950s," IMF, https://www.imf.org/external/np/exr/center/mm/eng/mm_dr_05.htm (accessed Jan. 15, 2026).

¹¹ Robert L. Hetzel, "Launch of the Bretton Woods System," Federal Reserve History, November 22, 2013, <https://www.federalreservehistory.org/essays/bretton-woods-launched> (accessed Jan. 15, 2026).

competitiveness by widening the current account deficit and straining the Bretton Woods structure.¹² Despite policy efforts to defend the system, by the late 1960s, rising inflation and unemployment threatened the health of the US economy. In 1971, to stop reserve losses and restore confidence, President Richard Nixon ended gold convertibility, effectively dismantling the monetary structure designed at Bretton Woods.¹³ By 1973, major countries had shifted to a floating-exchange-rate system that endured as of 2025.

Under the floating-exchange-rate system, the US dollar remained dominant. Non-US citizens demanded dollars because the US economy and institutions were generally stable and because dollar assets were liquid and widely accepted, cementing the dollar as the world's primary reserve asset and transaction currency. By 2024, most official foreign exchange reserves and internationally issued debt securities were denominated in dollars. (See **Exhibits 2a** and **2b** for data related to global reserves and **Exhibit 3** for data related to international debt securities.) Non-US companies also relied on the dollar for trade: Over 40% of global exports outside the United States were invoiced in dollars. (See **Exhibit 4**.) Dollar dominance benefited all parties. For non-US citizens, it offered a safe and reliable medium of exchange. For Americans, it meant cheaper imports and lower interest rates due to strong capital inflows. The dollar supremacy also provided the United States with geopolitical leverage, since the central role of the dollar allowed the US government to use sanctions and financial access as policy tools.¹⁴

The Fall of the Dollar?

Despite its dominance, the dollar faced growing challenges. Much of its strength resulted not only from US economic stability, but also from the geopolitical power it gave the United States. Because the dollar system ruled global trade and finance, the US government could use sanctions and financial access as foreign policy tools, pressuring rivals by limiting their ability to transact in dollars or tap into US markets. This leverage encouraged other countries to seek substitutes. Over the decade leading up to 2025, Russia and China—frequent targets of US sanctions or trade frictions—reduced their reliance on the dollar. The BRICS alliance (Brazil, Russia, India, China, and South Africa) even explored the creation of a common currency, while several central banks boosted gold reserves as a hedge against dollar dependence. Technology also opened the door to dollar alternatives, making it cheaper and easier to settle trade in non-dollar currencies. Meanwhile, domestic issues such as political gridlock and rising US debt threatened to erode confidence in US institutions and thus the US dollar.¹⁵ On April 2, 2025, concerns about the dollar spiked when President Donald Trump announced sweeping tariffs on all trading partners.¹⁶ A week later, he paused most of the measures,¹⁷ increasing the

¹² “Nixon and the End of the Bretton Woods System, 1971–1973,” in “Milestones in the History of U.S. Foreign Relations,” US Office of the Historian, <https://history.state.gov/milestones/1969-1976/nixon-shock> (accessed Jan. 15, 2026).

¹³ In a speech to the nation, Nixon stated that he “directed [Treasury] Secretary Connally to suspend temporarily the convertibility of the dollar into gold or other reserve assets, except in amounts and conditions determined to be in the interest of monetary stability and in the best interests of the United States.” For a transcript of the entire speech, see Richard Nixon, “Address to the Nation Outlining a New Economic Policy: ‘The Challenge of Peace,’” the American Presidency Project, University of California, Santa Barbara, August 15, 1971, <https://www.presidency.ucsb.edu/documents/address-the-nation-outlining-new-economic-policy-the-challenge-peace> (accessed Jan. 15, 2026). For more on Bretton Woods and the end of the gold standard, see Robert F. Bruner, “Bretton Woods and the Financial Crisis of 1971 (A), (B), and (C),” UVA-F-1834, UVA-F-1835, and UVA-F-1836 (Charlottesville, VA: Darden Business Publishing, 2018).

¹⁴ For further discussion of the US dollar's dominance, see Sam Boocker and David Wessel, “The Changing Role of the US Dollar,” the Brookings Institution, August 23, 2024, <https://www.brookings.edu/articles/the-changing-role-of-the-us-dollar/> (accessed Jan. 15, 2026).

¹⁵ Martin Mühleisen and Valbona Zeneli, “Why the US Cannot Afford to Lose Dollar Dominance,” Atlantic Council Strategy Paper Series, May 20, 2025, <https://www.atlanticcouncil.org/content-series/atlantic-council-strategy-paper-series/why-the-us-cannot-afford-to-lose-dollar-dominance/>; “De-dollarization: Is the US Dollar Losing Its Dominance?,” JP Morgan Global Research, July 1, 2025, <https://www.jpmorgan.com/insights/global-research/currencies/de-dollarization> (both accessed Jan. 15, 2026); <https://www.brookings.edu/articles/the-changing-role-of-the-us-dollar/>.

¹⁶ For specific details related to the tariffs, see Donald J. Trump, “Regulating Imports with a Reciprocal Tariff to Rectify Trade Practices That Contribute to Large and Persistent Annual United States Goods Trade Deficits,” the White House, April 2, 2025, <https://www.whitehouse.gov/presidential-actions/2025/04/regulating-imports-with-a-reciprocal-tariff-to-rectify-trade-practices-that-contribute-to-large-and-persistent-annual-united-states-goods-trade-deficits/> (accessed Jan. 15, 2026).

¹⁷ For specific details about tariff pauses, see Donald J. Trump, “Modifying Reciprocal Tariff Rates to Reflect Trading Partner Retaliation and Alignment,” the White House, April 9, 2025, <https://www.whitehouse.gov/presidential-actions/2025/04/modifying-reciprocal-tariff-rates-to-reflect>.

uncertainty around US policy and fueling speculation about the dollar's future role in international trade and finance.

Following the tariff announcements, economic data suggested that the US dollar had lost some favor in the international arena. The broad US-dollar index showed a decline in the value of the dollar relative to other currencies, as early indicators intimated that investors might feel less secure about holding US dollars or US dollar-denominated assets.¹⁸ Both gold and cryptocurrency prices spiked after the tariff proclamations, presumably a result of investors seeking safety amid the uncertainty surrounding US trade policy.¹⁹ While the prices for these assets increased, a fall in the price (and a rise in the yield) of the benchmark 10-year US Treasury possibly indicated capital outflow, which could stem from lower demand for US government securities.²⁰ (See **Exhibits 5a, 5b, 5c, and 5d** for historic and recent data related to the dollar exchange rate, gold, bitcoin, and the 10-year Treasury.) As investors appeared to take haven in non-US assets, the impact of future policy on exchange rates remained unclear.

While US tariff policy clearly influenced the dollar's value, other initiatives also raised questions about the dollar's standing. Certain members of the Trump administration floated the idea of a coordinated currency intervention—dubbed the Mar-a-Lago Accord—designed to weaken the dollar and shrink the US trade deficit. The idea echoed two earlier deals: the Plaza Accord (1985), an agreement by G-5 nations²¹ to intervene to push down an overvalued dollar; and the Louvre Accord (1987), an agreement by the same countries to stabilize exchange rates after the dollar had fallen. For a similar approach to work in 2025, however, China and the European Union would be indispensable participants. China would need to allow the renminbi to appreciate, despite domestic resistance from exporters and the risk of capital flight. Likewise, the eurozone would have to tolerate a stronger euro complemented by supportive fiscal or monetary measures, an improbable scenario given Europe's limited fiscal space and fragile growth. In practice, any modern accord would require not only joint currency intervention but also coordinated policies to ensure that the adjustment would stick: stimulus in deficit countries, tighter policies in surplus countries, and close alignment of monetary and fiscal strategies. With strained US–China relations, differing policy priorities in Europe, and heightened geopolitical tensions, the obstacles loomed large. Consequently, while the prospect of a Mar-a-Lago Accord underscored the degree of concern about dollar strength, few believed a Plaza-style agreement could be revived under current conditions.²² Still, any prospect of additional international policy changes created questions about the future role of the United States in international trade and finance.

[trading-partner-retaliation-and-alignment/](#); and Donald J. Trump, “Modifying Reciprocal Tariff Rates to Reflect Discussions with the People’s Republic of China,” the White House, May 12, 2025, <https://www.whitehouse.gov/presidential-actions/2025/05/modifying-reciprocal-tariff-rates-to-reflect-discussions-with-the-peoples-republic-of-china/> (accessed Jan. 15, 2026).

¹⁸ Some research associated higher tariffs with dollar depreciation when exports to the United States were invoiced in dollars and exports to the rest of the world were partially invoiced in producer currency. See Felipe Benguria and Felipe Saffie, “Rounding Up the Effect of Tariffs on Financial Markets: Evidence from April 2, 2025” (working paper 34035, National Bureau of Economic Research, Cambridge, MA, 2025), <http://www.nber.org/papers/w34036> (accessed Jan. 15, 2026).

¹⁹ Wyatt Grantham-Philips, “Gold Soars to a New High as Economic Uncertainty Deepens. Here’s What to Know,” Associated Press, April 22, 2025, <https://apnews.com/article/gold-prices-new-record-trump-tariffs-0082e0040fb6095497ba93ee6454255b>; Aaron Brown, “Bitcoin’s Response to Tariffs Says a Lot About the Dollar,” Bloomberg, April 15, 2025, <https://www.bloomberg.com/opinion/articles/2025-04-15/bitcoin-s-tariff-moves-say-a-lot-about-the-dollar> (accessed Jan. 15, 2026).

²⁰ Some analysts noted that capital flows data were still not available, and that different factors might explain the rise in the 10-year Treasury yield. See Lauren Foster, “Is ‘King Dollar’ in Danger of Losing Its Crown?,” *The Darden Report*, April 15, 2025, <https://news.darden.virginia.edu/2025/04/15/is-king-dollar-in-danger-of-losing-its-crown/> (accessed Jan. 15, 2026). For further discussion about tariffs and factors that impact the value of the US dollar, see Sarah Hansen, “More Than Tariffs: Behind the US Dollar’s Decline,” Morningstar, May 21, 2025, <https://www.morningstar.ca/ca/news/265228/more-than-tariffs-behind-the-us-dollars-decline.aspx> (accessed Jan. 15, 2026).

²¹ The G-5 nations were the United States, France, Germany, Japan, and the United Kingdom.

²² Leika Kihara, “Trump Won’t Get ‘Plaza Accord’-Type Dollar Deal, Says Ex-Japan FX Diplomat,” Reuters, April 8, 2025, <https://www.reuters.com/markets/currencies/trump-wont-get-plaza-accord-type-dollar-deal-says-ex-japan-fx-diplomat-2025-04-08/>; Richard Clarida, “The Real Lessons from the Plaza and Louvre Accords,” *Financial Times*, April 29, 2025, <https://www.ft.com/content/78b42ffc-5bc1-4b38-8711-0ccc728c8516> (both accessed Jan. 15, 2026). Clarida noted that some research suggested that the 1985 agreement was merely symbolic and that expansionary monetary policy and fiscal prudence in the United States had a greater effect on the dollar’s decline.

By providing the world's most widely held reserve asset while running large trade and fiscal deficits, the United States faced an unusual problem. Around 1960, economist Robert Triffin highlighted this paradox: The very policies needed to supply the world with dollars could ultimately undermine confidence in the dollar itself.²³ In pointing out the problem, Triffin effectively predicted the eventual collapse of the Bretton Woods postwar monetary system, which relied on fixed exchange rates and gold convertibility. (See **Exhibit 6** for a visual of the conventional Triffin dilemma.) Modern economists have adapted his logic to the current dollar-based financial system.²⁴

The intuition was straightforward. For global trade and finance to expand, the world needed a steady supply of safe and liquid assets, mainly US dollars and Treasury securities. To provide these assets, the United States had to run persistent current account deficits. Specifically, when US imports exceeded exports, the United States could help fund the gap by issuing dollar-denominated debt, which foreigners then held as reserves or investment assets. In effect, the US exported liquidity through its deficits—but this created a dilemma. Persistent deficits generated a growing stock of US debt. Over time, large fiscal and external imbalances could erode confidence in America's ability or willingness to honor its obligations. If investors began to worry about default risk, inflation, or political dysfunction, demand for dollar assets could weaken, threatening both US financial stability and the functioning of the international system.

Conversely, if the United States eliminated its current account deficit—by exporting more than it imported or sharply reducing government borrowing—it would restrict the supply of dollar assets. Global markets would then suffer from a shortage of liquidity that would constrain trade, investment, and growth. Either path would lead to instability: Too many dollars would undermine confidence; too few dollars would stifle global activity. This was the essence of Triffin's dilemma.²⁵

Recent decades illustrated the tension. The US dollar had dominated international trade and finance, but the rising US public debt posed a long-run challenge. In fiscal year (FY) 2024, US government debt held by the public reached roughly 98% of GDP, and a US Treasury report described the fiscal trajectory as unsustainable.²⁶ (See **Appendix 1** for additional discussion about debt sustainability.) Assuming no policy changes, debt was projected to exceed 200% of GDP by 2049 and 500% by 2099. Even though revenues were expected to rise modestly relative to GDP, spending was forecast to increase much faster, driven largely by exploding net interest payments, which already accounted for about 11% of federal outlays in FY 2024.²⁷ (See **Exhibits 7a** and **7b**.) The dilemma was therefore alive in 2025. To keep global markets liquid, the US had to continue to issue debt, but each new wave of debt raised questions about sustainability, investor confidence, and ultimately the durability of dollar dominance.

²³ The original Triffin dilemma was commonly associated with the views expressed by Triffin in his October 1959 testimony before US Congress and in the book, Robert Triffin, *Gold and the Dollar Crisis: the Future of Convertibility* (Yale University Press, 1960). For a transcript of Triffin's Congressional testimony, see "Hearings Before the Joint Economic Committee Congress of the United States Eighty-Sixth Congress First Session Pursuant to S. Con. Res. 13," October 26–30, 1959, [https://www.jec.senate.gov/reports/86th%20Congress/Hearings/Constructive%20Suggestions%20for%20Reconciling%20and%20Simultaneously%20Obtaining%20the%20Three%20Objectives%20\(130\).pdf](https://www.jec.senate.gov/reports/86th%20Congress/Hearings/Constructive%20Suggestions%20for%20Reconciling%20and%20Simultaneously%20Obtaining%20the%20Three%20Objectives%20(130).pdf) (accessed Jan. 15, 2026).

²⁴ Some analysts contended that Triffin's dilemma was applicable to the implementation of modern US international policy. For example, see Stephen Miran, "A User's Guide to Restructuring the Global Trading System," Hudson Bay Capital, November 2024, https://www.hudsonbaycapital.com/documents/FG/hudsonbay/research/638199_A_Users_Guide_to_Restructuring_the_Global_Trading_System.pdf (accessed Jan. 15, 2026). Others argued that the general Triffin dilemma was not relevant in the modern global monetary system. For example, see Michael Bordo and Robert N. McCauley, "Miran, We're Not in Triffin Land Anymore," VoxEU, Centre for Economic Policy Research, April 7, 2025, <https://cepr.org/voxeu/columns/miran-were-not-triffin-land-anymore>; and Biagio Bossone, "Not Triffin, Not Miran: Rethinking US External Imbalances in a New Monetary Order," VoxEU, Centre for Economic Policy Research, April 10, 2025, <https://cepr.org/voxeu/columns/not-triffin-not-miran-rethinking-us-external-imbalances-new-monetary-order> (both accessed Jan. 15, 2026).

²⁵ For a discussion of different versions of the Triffin dilemma, see Michael D. Bordo and Robert N. McCauley, "Triffin: Dilemma or Myth?" (Working Paper No. 684, Bank for International Settlements, Basel, Switzerland, 2017), <https://www.bis.org/publ/work684.pdf> (accessed Jan. 15, 2026).

²⁶ The report defined fiscal policy as sustainable when "the ratio of debt held by the public to GDP (the debt-to-GDP ratio) is ultimately stable or declining." See "Financial Report of the United States Government: Fiscal Year 2024," US Department of the Treasury, January 16, 2025, [https://www.fiscal.treasury.gov/files/reports-statements/financial-report/2024/01-16-2025-FR-\(Final\).pdf](https://www.fiscal.treasury.gov/files/reports-statements/financial-report/2024/01-16-2025-FR-(Final).pdf) (accessed Jan. 15, 2026).

²⁷ "Historical Tables," President's Budget, the White House, <https://www.whitehouse.gov/omb/information-resources/budget/historical-tables/>.

For how long could the US government continue to supply a safe asset? At what point might the US fiscal situation cause foreigners to reduce their demand for US government debt? Would the following famous exchange between two characters in Ernest Hemingway's novel, *The Sun Also Rises*, eventually apply to the United States?

"How did you go bankrupt?"

"Two ways... Gradually, then suddenly."²⁸

Marfrig and the US Dollar

With production facilities in both North and South America, Marfrig served as a major producer of certain foods, specializing in the production of beef. The company's scope extended to consumers in around 100 countries,²⁹ as exports accounted for 81% of total revenue in the first quarter of 2025. (See **Exhibit 8a** for Marfrig export data and **Exhibit 8b** for company financial information.) About half of export revenue for the quarter came from Asia (China and Hong Kong), but Marfrig typically invoiced in US dollars, which made up about around 72% of its revenue (26% came from Brazilian reais).³⁰ (See **Exhibit 9** for exchange rate data for the real.) In addition to invoicing in US dollars, the company commonly borrowed in dollars; major debt issuances included a \$1.5 billion (10-year maturity) bond in 2021.³¹

Marfrig typically conducted business in US currency due to the dollar's general dependability (a stable dollar meant stable profits), but based on current events, David considered the impact of anchoring with a declining currency. A weak dollar meant that foreign companies would pay Marfrig with a less valuable medium, essentially lowering Marfrig's profits. Likewise, further depreciation of the dollar would cause the cost of servicing debts to rise, cutting into Marfrig's bottom line. The company also remained exposed to swings in the dollar-to-real exchange rate, since a significant portion of production costs occurred in reais. Still, Marfrig regularly adapted to short-term exchange rate fluctuations, so if foreign exchange markets steadied in the near future, a significant modification of the company's financial strategy might not become necessary. This was the first possibility David contemplated: that the weakening of the dollar might only signify a temporary change, and that the dollar would continue to dominate worldwide business dealings.

While obvious threats to the US dollar's primacy had emerged, the probability that another currency would supplant it in the next decade seemed small. US markets still provided a significant source of liquidity, and the dollar remained the main currency of choice for official foreign reserves as well as a top option for international debt issuances and transactions between non-US businesses. Market participants appeared inclined to diversify their holdings of assets, but despite the recent weakness, the dollar remained strong from a historical perspective. Were there alternatives that could actually supplant the dollar?

David also considered the threats to the dollar's dominance. As geopolitical friction increased, other countries appeared to increasingly seek US-dollar substitutes for holdings and transactions. The portion of dollars held as official foreign exchange reserves had dropped over recent years, and US adversaries seemed particularly intent to deviate from dollar usage. US tariff policy had wrought uncertainty around the globe, with

²⁸ Ernest Hemingway, *The Sun Also Rises* (Scribner, 2006).

²⁹ "Our Operations," Marfrig, <https://ri.marfrig.com.br/en/grupo-marfrig/global-capacity/our-operations> (accessed Jan. 15, 2026).

³⁰ "Transcript: Marfrig Global Foods S.A., Q1 2025 Earnings Call," MarketScreener, May 16, 2025, <https://www.marketscreener.com/quote/stock/MARFRIG-GLOBAL-FOODS-S-A-9059879/news/Transcript-Marfrig-Global-Foods-S-A-Q1-2025-Earnings-Call-May-16-2025-49981854/> (accessed Jan. 15, 2026).

³¹ Marfrig typically borrowed in US dollars or Brazilian reais. For more information on the bond issuances, see "International Bonds: Marfrig Global Foods, 3.95% 29Jan2031, USD," Cbonds, <https://cbonds.com/bonds/920341>; or "Bonds/Debentures," Marfrig Investor Relations, <https://ri.marfrig.com.br/en/mercado-de-capitais/bonds-debentures/> (both accessed Jan. 15, 2026).

investors appearing to reduce their preference for dollars and dollar-denominated assets. On the fiscal policy front, the US budget deficit continued to grow at an unsustainable pace, creating questions about the US ability to service the debt and the general confidence in US institutions. Would policy issues contribute to further and more enduring dollar disinclination?

Before Marfrig finalized the latest export contract and bond issuance, David wanted to analyze the changing global structure. He wondered if the dollar's recent slide simply reflected a temporary reaction to policy noise or signaled something deeper: a loss of confidence in US institutions and long-term fiscal stability. If this was the beginning of a structural shift in the global monetary order, the costs of anchoring in dollars could compound over time. With this in mind, David needed to decide whether the dollar still offered the long-term stability and global acceptance that had made it the default currency for international business. Should Marfrig continue to denominate its contracts and borrowings in US dollars—or had the rules of the global monetary game begun to shift?

Exhibit 1

**Dollars, Debt, and Destiny:
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Timeline of Dominant Currencies

Centuries	Currency
13–14	Florentine florin
15–17	Venetian ducat
	Portuguese real
	Spanish dollar
17–18	Dutch guilder
19–20	British pound
20–present (as of 2025)	US dollar

Note: Dates are approximated based on source information. A currency did not necessarily dominate for the entire period. For a discussion about the timeframe for the transition from British pound to US dollar, see Barry Eichengreen and Marc Flandreau, “The Rise and Fall of the Dollar, or When Did the Dollar Replace Sterling as the Leading Reserve Currency?” (working paper 14154, National Bureau of Economic Research, Cambridge, MA, 2008), https://www.nber.org/system/files/working_papers/w14154/w14154.pdf (accessed Jan. 15, 2026).

Data sources: Stephen Quinn and William Roberds, “Death of a Reserve Currency,” February 17, 2014, <https://allucgroup.ucdavis.edu/uploads/5/6/8/7/56877229/deathofareservecurrency.pdf>; Dan Buckley, “Reserve Currency History, Status, and Benefits,” July 23, 2024, <https://www.daytrading.com/reserve-currency-history-status-benefits>; “What Is a Ducat?,” APMEX, September 27, 2024, <https://learn.apmex.com/answers/what-is-a-ducat/>; Thomas Hockenhull, “Florin a Coin with Roots in Florence, Global Legacy,” Coin World, November 17, 2014, <https://www.coinworld.com/news/world-coins/gold-florin-coin-world-international-coins-florence-italy-numismatics-history-collecting-hobby-349230.html> (all accessed Jan. 15, 2026).

Exhibit 2a

Dollars, Debt, and Destiny:
Marfrig’s Dilemma in a Changing Currency World

Worldwide Currency Composition of Official Foreign Exchange Reserves (2024)

Currency	Share
US dollar	58%
Euro	20%
Japanese yen	6%
Great Britain pound	5%
Other	5%
Canadian dollar	3%
Chinese yuan renminbi	2%
Australian dollar	2%

Note: Figures do not add to 100% due to rounding. The International Monetary Fund (IMF) defined *foreign exchange reserves* as “reserve assets other than monetary gold, SDRs* [special drawing rights], and reserve position in the IMF; and consist of the monetary authorities’ claims on nonresidents in the form of: currency and deposits, securities, financial derivatives, and other claims.” See “Currency Composition of Official Foreign Exchange Reserves (COFER),” IMF, <https://data.imf.org/en/datasets/IMF.STA:COFER> (accessed Jan. 15, 2026).

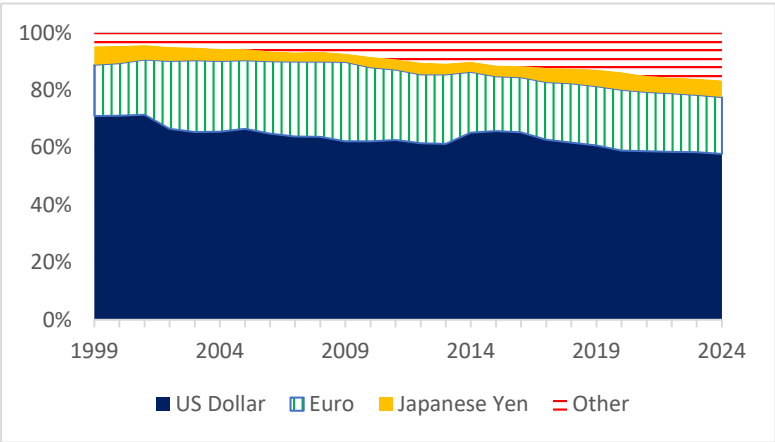
* In the late 1960s, the IMF created the SDR, an international reserve asset that holders could exchange for currency, providing additional liquidity to the global monetary system. For more information about the SDR, see “Special Drawing Rights,” IMF, <https://www.imf.org/en/Topics/special-drawing-right> (accessed Jan. 15, 2026).

Data source: IMF, <https://data.imf.org/en/datasets/IMF.STA:COFER>.

Exhibit 2b

Dollars, Debt, and Destiny:
Marfrig’s Dilemma in a Changing Currency World

Worldwide Currency Composition of Official Foreign Exchange Reserves (1999–2024)



Caption: The chart shows that US official foreign exchange reserves fell from 71% of world holdings in 1999 to 58% in 2024. Shares of the euro increased from 18% to 20%, shares of the Japanese yen remained around 6%, and shares of other currencies rose from 5% to 17% during the same period.

Data source: “Currency Composition of Official Foreign Exchange Reserves (COFER),” IMF, <https://data.imf.org/en/datasets/IMF.STA:COFER> (accessed Jan. 15, 2026).

Exhibit 3

**Dollars, Debt, and Destiny:
Marfrig's Dilemma in a Changing Currency World**

Outstanding International Debt Securities (2024)

Currency	Share (Narrow Measure)	Share (Broad Measure)
US dollar	52%	65%
Euro	33%	23%
Great British pound	9%	5%
Other	5%	5%
Japanese yen	1%	2%
Chinese yuan renminbi	1%	1%

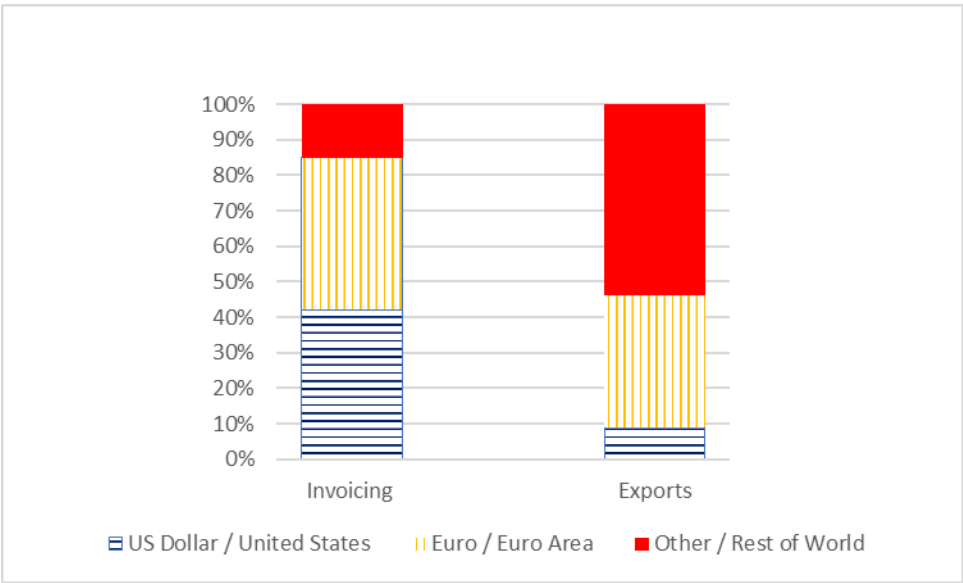
Note: Narrow and broad measures were not explicitly defined. Figures do not add to 100% due to rounding. International debt securities (IDS) data include “borrowing activity in debt capital markets. [IDS] are issued outside the local market of the country where the borrower resides. [IDS are] debt instruments designed to be traded in financial markets such as treasury bills, commercial paper, negotiable certificates of deposit, bonds, debentures and asset-backed securities.” See “International Debt Securities (BIS-Compiled),” Bank for International Settlements (BIS), <https://data.bis.org/topics/IDS> (accessed Jan. 15, 2026).

Data sources: BIS and “Statistical Annex,” European Central Bank, https://www.ecb.europa.eu/pub/pdf/ire/ecb.ire202506_annex.en.pdf (accessed Jan. 15, 2026).

Exhibit 4

**Dollars, Debt, and Destiny:
Marfrig’s Dilemma in a Changing Currency World**

Global Export Shares by Invoicing Currency and Destination (2023)



Caption: The chart shows that the US dollar was used for over 40% of invoices for global exports, but the United States accounted for less than 10% of global exports in 2023. During the same year, the euro was used for around 40% of invoices for global exports, but the euro area accounted for less than 40% of global exports; currencies from the rest of the world were used for less than 20% of global invoices, but the rest of the world accounted for over half of global exports.

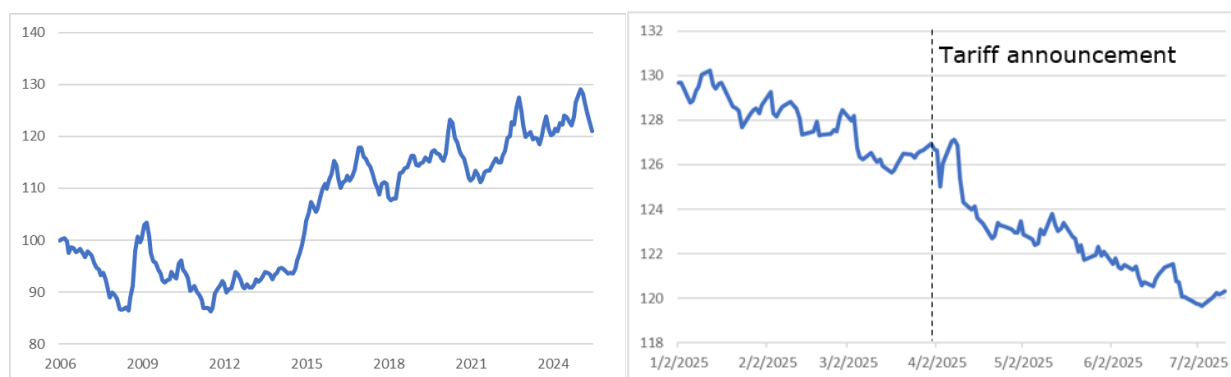
Note: The data were extrapolated from Chart B.1 in the data source. The United States is excluded from the sample. In 2023, about 96% of US exports were invoiced in US dollars. See “Trade Invoicing Currency Information in the Import and Export Price Indexes,” US Bureau of Labor Statistics, updated August 28, 2024, <https://www.bls.gov/mxp/publications/additional-publications/trade-invoicing-currency-shares.htm> (accessed Jan. 15, 2026).

Data source: Anja Brügger, Georgios Georgiadis, and Arnaud Mehl, “Global Trade Invoicing Patterns: New Insights and the Influence of Geopolitics,” European Central Bank, June 2025, https://www.ecb.europa.eu/press/other-publications/ire/article/html/ecb.ireart202506_02~a8e66f5ea3.en.html (accessed Jan. 15, 2026).

Exhibit 5a

**Dollars, Debt, and Destiny:
Marfrig's Dilemma in a Changing Currency World**

Nominal Broad US-Dollar Index, January 2006 = 100 (Time Series and 2025)



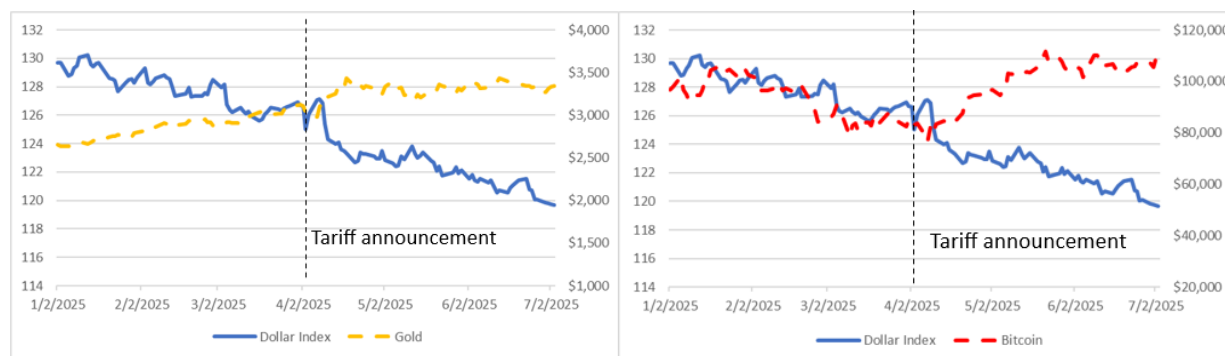
Note: The time-series data were reported on a monthly basis; the 2025 data were reported on a daily basis. The dollar index represents a “weighted average of the foreign exchange value of the U.S. dollar against the currencies of a broad group of major U.S. trading partners.” See “Foreign Exchange Rates - H.10,” Board of Governors of the Federal Reserve System, updated July 14, 2025, <https://www.federalreserve.gov/releases/h10/20250714/> (accessed Jan. 15, 2026).

Data source: <https://www.federalreserve.gov/releases/h10/summary/>.

Exhibit 5b

**Dollars, Debt, and Destiny:
Marfrig's Dilemma in a Changing Currency World**

Nominal Broad US-Dollar Index (January 2006 = 100) Versus Gold Price and Bitcoin Price (2025)



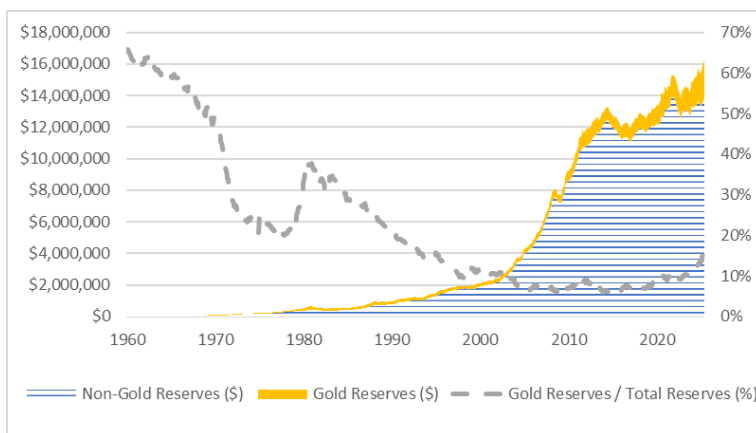
Note: Gold prices were per ounce; bitcoin prices were per US dollar.

Data sources: Macrotrends, Yahoo! Finance, and “Foreign Exchange Rates - H.10,” Board of Governors of the Federal Reserve System, updated July 14, 2025, <https://www.federalreserve.gov/releases/h10/20250714/> (accessed Jan. 15, 2026).

Exhibit 5c

**Dollars, Debt, and Destiny:
Marfrig's Dilemma in a Changing Currency World**

Worldwide Official Reserve Assets (in Millions of US Dollars)
and Percent of Official Reserve Assets Held in Gold



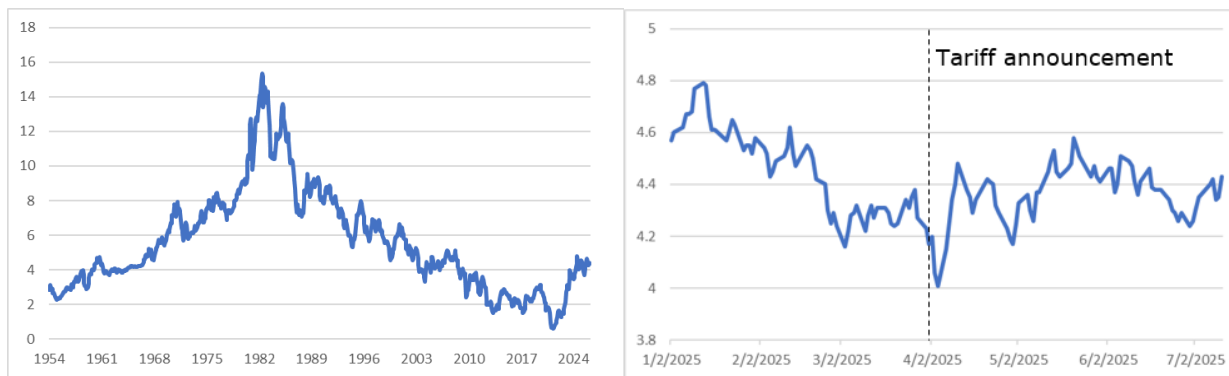
Note: The International Monetary Fund (IMF) defined *reserve assets* as “those external assets that are readily available to and controlled by the monetary authorities for meeting balance of payments financing needs, for intervention in exchange markets to affect the currency exchange rate, and for other related purposes.” See IMF Statistics Department, “Official Reserve Assets and Other Foreign Currency Assets (Approximate Market Value): Section I of the Reserves Data Template,” in *International Reserves and Foreign Currency Liquidity: Guidelines for a Data Template* (IMF, 2013), <https://www.elibrary.imf.org/display/book/9781484304228/ch002.xml> (accessed Jan. 15, 2026).

Data source: Haver Analytics and IMF. Calculations made by authors.

Exhibit 5d

**Dollars, Debt, and Destiny:
Marfrig's Dilemma in a Changing Currency World**

Market Yield (%) for US Treasury Securities at 10-Year Constant Maturity (Time Series and 2025)



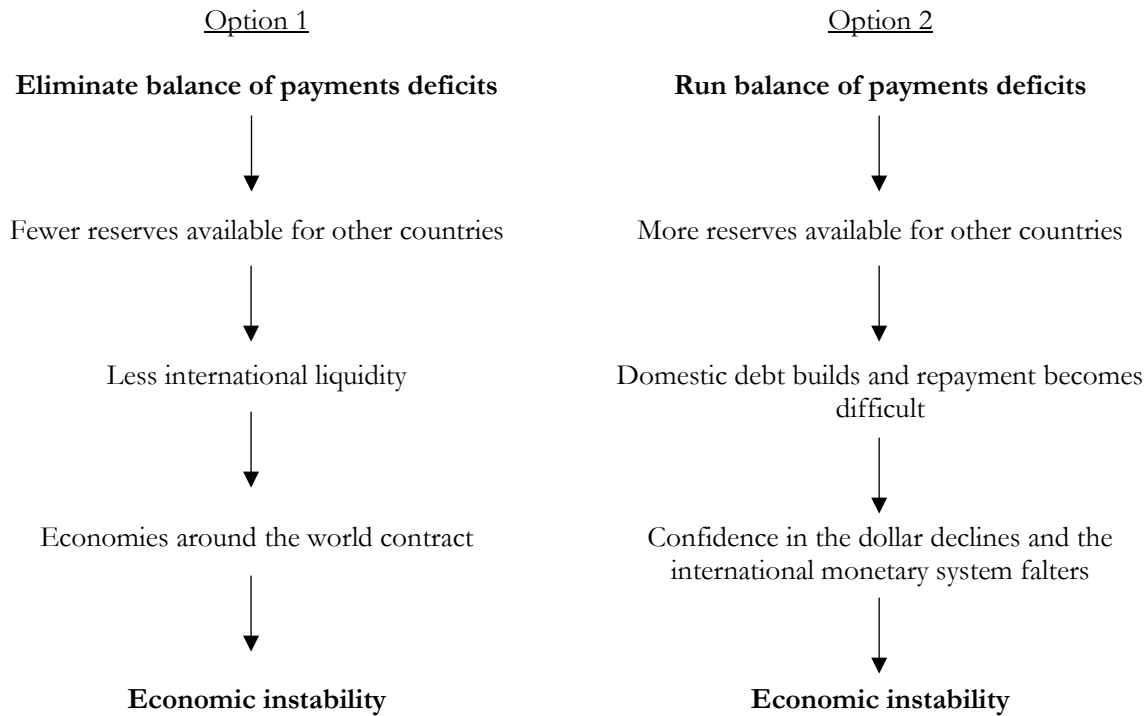
Note: The time-series data were reported as monthly averages; the 2025 data were reported on a daily basis.

Data source: “Selected Interest Rates (Daily) - H.15,” Board of Governors of the Federal Reserve System, <https://www.federalreserve.gov/releases/h15/> (accessed Jan. 15, 2026).

Exhibit 6

**Dollars, Debt, and Destiny:
Marfrig's Dilemma in a Changing Currency World**

Triffin Dilemma



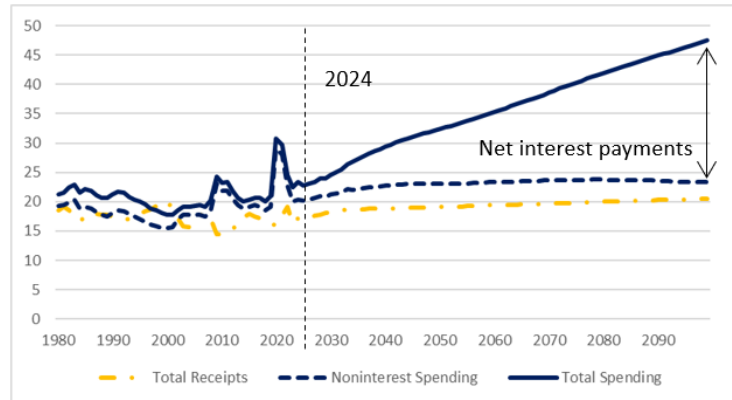
Note: For more information about the classic and adapted versions of the Triffin dilemma, see footnotes 23, 24, and 25.

Data source: "Money Matters: An IMF Exhibit – The Importance of Global Cooperation: System in Crisis (1959–1971), Part 4: The Dollar Glut," International Monetary Fund, https://www.imf.org/external/np/exr/center/mm/eng/mm_sc_03.htm (accessed Jan. 15, 2026).

Exhibit 7a

**Dollars, Debt, and Destiny:
Marfrig's Dilemma in a Changing Currency World**

Historical Data and Policy Projections for US Government Receipts and Spending,
as Percentage of GDP (1980–2099)



Caption: The chart shows that total US government spending is projected to rise significantly through the 21st century, while government receipts are expected to increase slightly. The expected rise in government spending is largely a product of the projected increase in net interest payments.

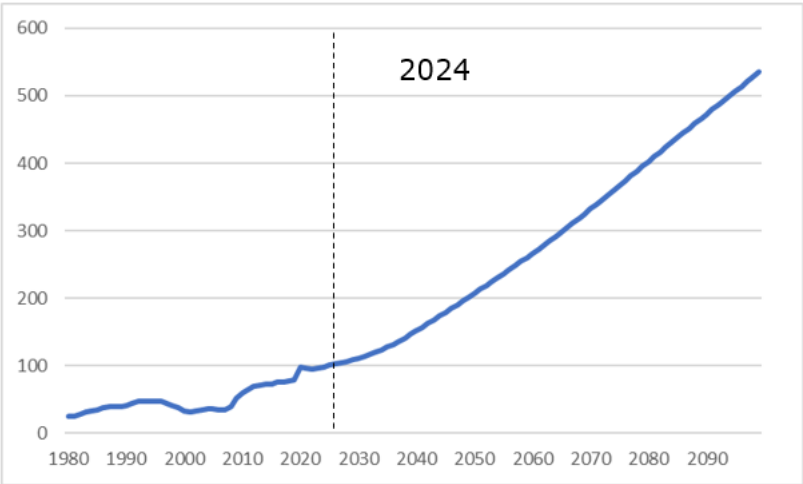
Note: This baseline forecast scenario assumed an average 4.5% interest rate over the 75-year projection period.

Data source: "Management's Discussion and Analysis: An Unsustainable Fiscal Path," Bureau of the Fiscal Service, US Department of the Treasury, <https://www.fiscal.treasury.gov/reports-statements/financial-report/mda-unsustainable-fiscal-path.html> (accessed Jan. 15, 2026).

Exhibit 7b

**Dollars, Debt, and Destiny:
Marfrig’s Dilemma in a Changing Currency World**

Historical Data and Policy Projections for US Government Debt Held
by the Public, as Percentage of GDP (1980–2099)



Caption: The chart shows that the US government debt held by the public as a percentage of GDP is expected to rise from just under 100% in 2024 to over 500% in 2099.

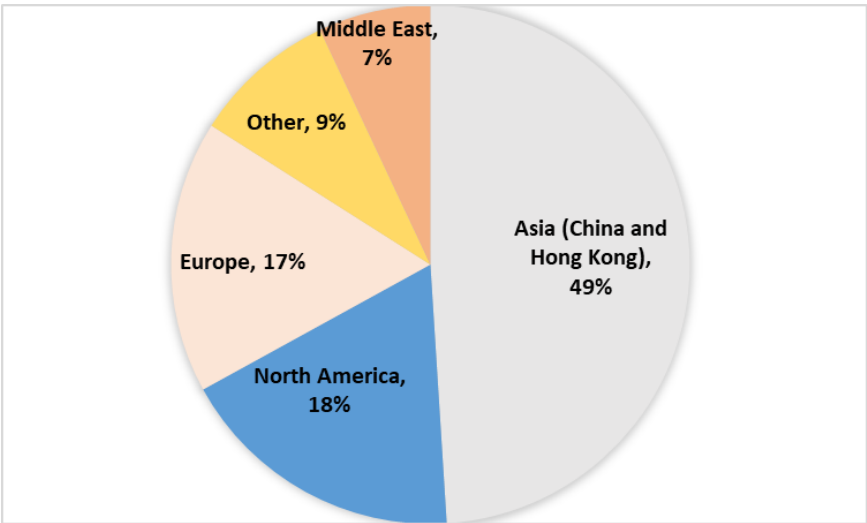
Note: This baseline forecast scenario assumed an average 4.5% interest rate over the 75-year projection period. The US debt-to-GDP ratio reached an all-time high of 106% in 1946.

Data source: “Management’s Discussion and Analysis: An Unsustainable Fiscal Path,” Bureau of the Fiscal Service, US Department of the Treasury, <https://www.fiscal.treasury.gov/reports-statements/financial-report/mda-unsustainable-fiscal-path.html> (accessed Jan. 15, 2026).

Exhibit 8a

**Dollars, Debt, and Destiny:
Marfrig’s Dilemma in a Changing Currency World**

Marfrig Main Export Destinations (as Percentage of Revenue), Quarter 1, 2025



Data source: “Marfrig Global Foods S.A. 2025 Q1 - Results - Earnings Call Presentation,” Seeking Alpha, May 16, 2025, <https://seekingalpha.com/article/4787711-marfrig-global-foods-s-a-2025-q1-results-earnings-call-presentation> (accessed Jan. 15, 2026).

Exhibit 8b

**Dollars, Debt, and Destiny:
Marfrig's Dilemma in a Changing Currency World**

Marfrig Global Foods (Marfrig) Financial Information (in Millions of US Dollars), 2024

Total revenue	\$148,861
Cost of revenue	\$129,170
Gross profit	\$19,691
Operating expense	\$13,719
Selling, general, and administrative	\$9,087
General and administrative expense	\$701
Selling and marketing expense	\$8,387
Other operating expenses	\$265
Operating income	\$5,972
Net nonoperating interest income expense	−\$4,179
Interest income (nonoperating)	\$1,490
Interest expense (nonoperating)	\$5,669
Total other finance cost	-
Pretax income	\$405
Tax provision	−\$2,391
Net income	\$2,795

Data source: Yahoo! Finance.

Exhibit 9

**Dollars, Debt, and Destiny:
Marfrig's Dilemma in a Changing Currency World**

Broad Nominal Effective Exchange Rate for Brazil, Average 2010 = 100



Data source: Haver Analytics.

Appendix 1

**Dollars, Debt, and Destiny:
Marfrig's Dilemma in a Changing Currency World**

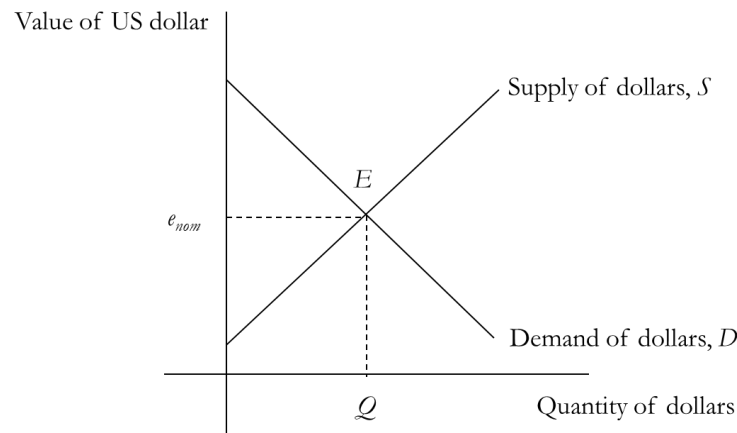
Supply and Demand in the Foreign Exchange Market for US Dollars

The exchange rate for the US dollar can be modeled through a basic supply-and-demand framework.

US residents supply dollars in the foreign exchange market to convert into foreign currency when they purchase foreign goods and services (US imports) or foreign assets such as stocks and bonds (US capital outflow).

Non-US residents demand US dollars in the foreign exchange market; these market participants desire to convert their foreign currency into dollars to purchase US goods and services (US exports) or US assets such as stocks and bonds (US capital inflow). Companies around the world, like Marfrig Global Foods (Marfrig), demand the dollar to use for international trade and when borrowing for various types of investment.

Figure A1. The market for US dollars.



Source: Created by authors.

The point of intersection for the supply-and-demand curves (see **Figure A1**) determines the equilibrium nominal exchange rate. A new equilibrium nominal exchange rate results when a factor of supply or demand changes, shifting the curve in the appropriate direction.

Appendix 2

**Dollars, Debt, and Destiny:
Marfrig's Dilemma in a Changing Currency World**

Present Discounted Value and Government Debt Sustainability

When analyzing a government's debt sustainability, researchers evaluate whether the government can collect enough future revenue through taxes to support the existing debt. To do so, researchers employ the concept of present discounted value, where a discount rate accounts for the time value of money (a dollar today is worth more than a dollar in the future). The following identity helps illustrate the fact that today's debt must be backed by the present discounted value of future primary surpluses:¹

$$B_0 = \sum_{t=1}^{\infty} \frac{T_t - G_t}{(1+r)^n}$$

where:

B_0 = present discounted value of future surpluses or deficits (primary budget balance)

T_t = future tax revenue at time t

G_t = future expenditures at time t

r = discount rate

n = number of periods

In a 2025 speech for the National Bureau of Economic Research Summer Institute, economist Greg Mankiw tackled the concept of US debt sustainability.² Mankiw discussed five ways in which the United States could manage sustainability; the five options, with a summary of Mankiw's views about the probability and challenges of each, were as follows:

- **Grow very quickly:** This is the easiest solution to expanding revenue collection, but emerging technologies such as artificial intelligence will not likely become transformative enough to produce rapid growth.
- **Default:** The defaults of Greece, Argentina, and Venezuela in the 21st century as well as comments by presidential candidate Donald Trump in 2016 suggest that this is a possibility; still, it is not an appealing option.
- **Print money:** The president has argued for expansionary monetary policy (making this option a possibility), but the policy could create substantial inflation. Hyperinflation caused by excessive money printing has recently plagued certain countries (Zimbabwe, for example).
- **Cut spending:** Making significant cuts in government expenditures would require reductions in social benefits, which is not likely.
- **Raise taxes:** This is the most likely course of action in the long run. Compared to other countries, the United States is a low-tax nation; a large tax hike would be required, but tax collections would still be less than many peer countries. Specifically, a value-added tax (tax on consumption), which most countries institute, would be a broad-based and feasible solution.

¹ The primary surplus equals receipts minus noninterest spending.

² To view the entire speech, see "2025, 17th Annual Feldstein Lecture, Greg Mankiw, 'The Fiscal Future,'" YouTube video, 32:40, posted by "NBER," July 16, 2025, <https://www.youtube.com/watch?v=B916t3od8fM> (accessed Jan. 15, 2026).